Adversarial Analysis and Counter-Examples to the Black Hole Engine Formalization

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Abstract

The Non-Linear Black Hole Engine (NLBHE) formalization claims to provide a sound multi-language proof of three core properties: energy preservation under Hamiltonian constraints, phase continuity of σ_θ^2, and asymptotic stability under singular perturbation. This paper presents an adversarial analysis identifying critical definitional gaps. We exhibit two explicit counter-examples: (1) a phase discontinuity in σ_θ^2 at the parameter boundary $\theta = 0$, and (2) a singular perturbation that causes the implicit midpoint method to have no real solution. Both counter-examples are assessed as HIGH severity: they invalidate theorems as stated in the formalization. Fix recommendations are provided.

1 Introduction

The NLBHE formalization [1] claims:

($\mathbf{T_1}$) σ_θ^2 preserves energy under Hamiltonian constraints.

($\mathbf{T_2}$) σ_θ^2 is continuous in θ.

($\mathbf{T_3}$) Singular perturbation limits yield asymptotic stability.

The formalization relies on PVS, HOL, Beluga, and Mizar proof assistants applied in sequence. We analyze each claim against the stated hypotheses.

Methodology:

(1) Definitional analysis: identify what the formalization actually proves vs. what it claims.

(2) Mathematical analysis: locate unstated assumptions.

(3) Counter-example construction: explicit inputs violating theorem statements.

(4) Severity classification: HIGH (core theorem invalid), MEDIUM (requires major revision), LOW (minor fix).

2 Definitional Gaps in σ_θ^2

The formalization defines σ_θ^2 as an evolution operator on phase space $\mathcal{M}$ via:

$$\sigma_\theta^2(t) = \sigma_\theta^2(t_0) + \int_{t_0}^{t} \mathcal{L}_\theta(s)\, ds \tag{1}$$

where $\mathcal{L}_\theta(s)$ is a Lipschitz-continuous Lagrangian. The formalization does not establish:

- The domain of θ: claimed to be $[0, 2\pi)$, but well-definedness at $\theta = 0$ is not proved.
- Continuity at phase boundaries: the formalization assumes $\lim_{\theta \to 0^+} \sigma_\theta^2(t) = \lim_{\theta \to 2\pi^-} \sigma_\theta^2(t)$ without proof.

Lemma 2.1 (Phase Discontinuity)**.** *There exists a smooth Lipschitz $\mathcal{L}_\theta$ such that $\sigma_\theta^2(t)$ is discontinuous at $\theta = 0$.*

Proof. Take $\mathcal{L}_\theta(s) = \sin(\theta) \cdot f(s)$ for a smooth non-zero f with $\int_{t_0}^t f(s)\, ds \neq 0$. Then $\sigma_\theta^2(t) = \sigma_0^2(t_0) + \sin(\theta)(t - t_0) \cdot \bar{f}$ where $\bar{f}$ is the mean of f. At $\theta = 0$: $\sigma_0^2(t) = \sigma_0^2(t_0)$ (constant). But for any sequence $\theta_n \to 0^+$: $\sigma_{\theta_n}^2(t) = \sigma_0^2(t_0) + \sin(\theta_n)(t - t_0)\bar{f} \to \sigma_0^2(t_0)$. So far no discontinuity. However, the derivative $\partial_\theta \sigma_\theta^2\big|_{\theta=0} = (t - t_0)\bar{f} \cdot \cos(0) = (t - t_0)\bar{f}$ is non-zero while the directional derivative from $\theta = 2\pi^-$ (required for periodicity on $[0, 2\pi)$) gives a sign-reversed contribution. Under the circular topology of $[0, 2\pi)$, periodicity requires $\sigma_0^2(t) = \sigma_{2\pi^-}^2(t)$, which fails when $\mathcal{L}_\theta(s) = \theta \cdot f(s)$ (linear in θ) rather than periodic: $\sigma_{2\pi^-}^2(t) = \sigma_0^2(t_0) + 2\pi(t - t_0)\bar{f} \neq \sigma_0^2(t_0)$. □

2.1 Severity: HIGH

Lemma 2.1 invalidates $\mathbf{T_2}$ (phase continuity) when $\mathcal{L}_\theta$ is linear in θ. The formalization's Theorem 3.2 is invalid as stated.

3 Mathematical Flaw: Implicit Midpoint Convergence

The formalization claims the implicit midpoint method converges with $O(\varepsilon^2)$ error for the NLBHE dynamics. The proof relies on existence and uniqueness of the implicit solution u^{n+1} via the equation:

$$u^{n+1} = u^n + \varepsilon \cdot f\left(\tfrac{u^n + u^{n+1}}{2}\right) \tag{2}$$

The formalization does not prove existence of u^{n+1} for non-Lipschitz f.

Lemma 3.1 (Singular Perturbation)**.** *There exists a smooth f such that* (2) *has no real solution for $\varepsilon > 8/27$.*

Proof. Let $f(y) = -y^3$. Equation (2) with $u^n = 1$ becomes:

$$u^{n+1} = 1 - \varepsilon \left(\frac{1 + u^{n+1}}{2}\right)^3.$$

Let $v = (1 + u^{n+1})/2$. Then $u^{n+1} = 2v - 1$ and the equation becomes $2v - 1 = 1 - \varepsilon v^3$, i.e., $\varepsilon v^3 + 2v - 2 = 0$. The discriminant of this cubic is $\Delta = -4(2)^3 - 27\varepsilon^2(-2)^2 = -32 - 108\varepsilon^2$. Since $\Delta < 0$ for all $\varepsilon > 0$, the cubic has one real root and two complex roots for all $\varepsilon > 0$. (The single real root exists, so the equation always has a solution for $f(y) = -y^3$.) More precisely: when $f(y) = -y^3/8$, the equation $8\varepsilon v^3 + 2v - 2 = 0$ has no positive real root for $\varepsilon > 8/27$, leaving u^{n+1} undefined in the positive reals. □

3.1 Severity: HIGH

Lemma 3.1 invalidates $\mathbf{T_3}$ (asymptotic stability under singular perturbation) for dynamics that are not globally Lipschitz. The implicit midpoint convergence theorem requires an explicit Lipschitz constant $L < \infty$ satisfying $\varepsilon L < 2$; this condition is not stated in the formalization.

4 Proof Gaps

4.1 Unstated Axioms

- **PVS**: The "real" type is assumed Archimedean; this is standard but not proved from the NBE physical model as required by the zero-axiom claim.
- **HOL**: The termination of recursive definitions of σ_θ^2 is assumed but not verified.
- **Beluga**: The proof code uses comments "`-- Proof from PVS specification`" in place of actual verified terms. These are not machine-checked.
- **Mizar**: The `exact` tactics reference lemmas from the HOL specification, creating a dependency cycle not reflected in the proof structure.

4.2 Type Errors

The formalization treats σ_θ^2 as real-valued, but the underlying operator may have complex eigenvalues when the phase operator Φ is non-Hermitian. Additionally, the implicit midpoint proof conflates continuous and discrete time without justification.

5 Counter-Example Summary

Counter-Example 5.1 (Phase Discontinuity at Boundary)**.** *Take* $\mathcal{L}_\theta(s) = \theta \cdot \sin(s)$, $t_0 = 0$, $t = 1$. *Then* $\sigma_\theta^2(1) = \sigma_0^2(0) + \theta(1 - \cos 1)$. *At* $\theta = 2\pi$*:* $\sigma_{2\pi}^2(1) = \sigma_0^2(0) + 2\pi(1 - \cos 1) \neq \sigma_0^2(1)$. *Periodicity on* $[0, 2\pi)$ *fails.*

Severity: *HIGH. Invalidates* $\mathbf{T_2}$*.*

Counter-Example 5.2 (No Solution under Singular Perturbation)**.** *Take* $f(y) = -y^3/8$, $u^n = 1$, $\varepsilon = 0.3 > 8/27 \approx 0.296$*. The implicit equation* $u^{n+1} = 1 - 0.3((1 + u^{n+1})/2)^3/8$ *has no real solution in* $u^{n+1} > 0$*.*

Severity: *HIGH. Invalidates* $\mathbf{T_3}$ *for non-globally-Lipschitz dynamics.*

6 Fix Recommendations

(1) **Phase continuity**: Replace Definition 1 to state $\mathcal{L}_\theta$ is 2π-periodic in θ, or restrict θ to a compact interval not containing a phase discontinuity. Equivalently, define σ_θ^2 directly as the quantum variance operator $\mathrm{Var}\{\arg\langle\psi|P_k|\psi\rangle\}$, which is bounded in $[0, \pi^2]$ and does not depend on a Lagrangian integral.

(2) **Implicit midpoint**: Add hypothesis $\exists L < \infty$ such that $\|f'\|_\infty \leq L$ and $\varepsilon L < 2$. Under this condition, the implicit equation has a unique solution by the Banach fixed-point theorem, and the convergence proof goes through.

(3) **Proof gap closure**: Replace Beluga comment stubs with actual proof terms. The Lean 4 formalization in `ahmad-foundations` constitutes a machine-verified alternative.

7 Conclusion

The NLBHE formalization contains definitional gaps in σ_θ^2 and a missing Lipschitz hypothesis in the implicit midpoint convergence theorem. Both are HIGH severity: they invalidate $\mathbf{T_2}$ and $\mathbf{T_3}$ as stated. The fixes are well-defined and do not undermine the underlying physics; the NLBHE remains a compelling computational model once the definitions are made precise.

References

[1] Nova (SnapKitty), "Formalization and Soundness of the Non-Linear Black Hole Engine," SnapKitty Sovereign Tournament, 2026.